

% ----- Facts -----

male(john).

male(paul).

male(mike).

male(david).

male(james).

male(robert).

male(tom).

female(mary).

female(susan).

female(lisa).

female(anna).

female(emily).

female(karen).

female(jane).

% parent(Parent, Child)

parent(john, paul).

parent(mary, paul).

parent(john, lisa).
parent(mary, lisa).

parent(paul, david).
parent(susan, david).

parent(paul, anna).
parent(susan, anna).

parent(mike, emily).
parent(karen, emily).

parent(mike, james).
parent(karen, james).

parent(david, tom).
parent(anna, jane).

% ----- Rules -----

% Father

```
father(X, Y) :-  
    male(X),  
    parent(X, Y).
```

% Mother

```
mother(X, Y) :-  
    female(X),  
    parent(X, Y).
```

% Grandfather

```
grandfather(X, Y) :-  
    male(X),  
    parent(X, Z),  
    parent(Z, Y).
```

% Grandmother

```
grandmother(X, Y) :-  
    female(X),  
    parent(X, Z),
```

parent(Z, Y).

% Brother

brother(X, Y) :-
 male(X),
 parent(P, X),
 parent(P, Y),
 X \= Y.

% Sister

sister(X, Y) :-
 female(X),
 parent(P, X),
 parent(P, Y),
 X \= Y.

% Uncle

uncle(X, Y) :-
 male(X),
 parent(P, Y),
 brother(X, P).

% Aunt

```
aunt(X, Y) :-  
    female(X),  
    parent(P, Y),  
    sister(X, P).
```

% Nephew

```
nephew(X, Y) :-  
    male(X),  
    (uncle(Y, X) ; aunt(Y, X)).
```

% Niece

```
niece(X, Y) :-  
    female(X),  
    (uncle(Y, X) ; aunt(Y, X)).
```

% Cousin

```
cousin(X, Y) :-  
    parent(P1, X),  
    parent(P2, Y),
```

(brother(P1, P2) ; sister(P1, P2)),
 $X \neq Y$.

Queries:

?- father(john, paul).
true.

?- mother(mary, lisa).
true.

?- grandfather(john, david).
true.

?- grandmother(mary, anna).
true.

?- brother(lisa, paul).
false.

?- sister(lisa, paul).

true.

?- uncle(mike, tom).

true.

?- aunt(karen, tom).

true.

?- cousin(david, jane).

true.